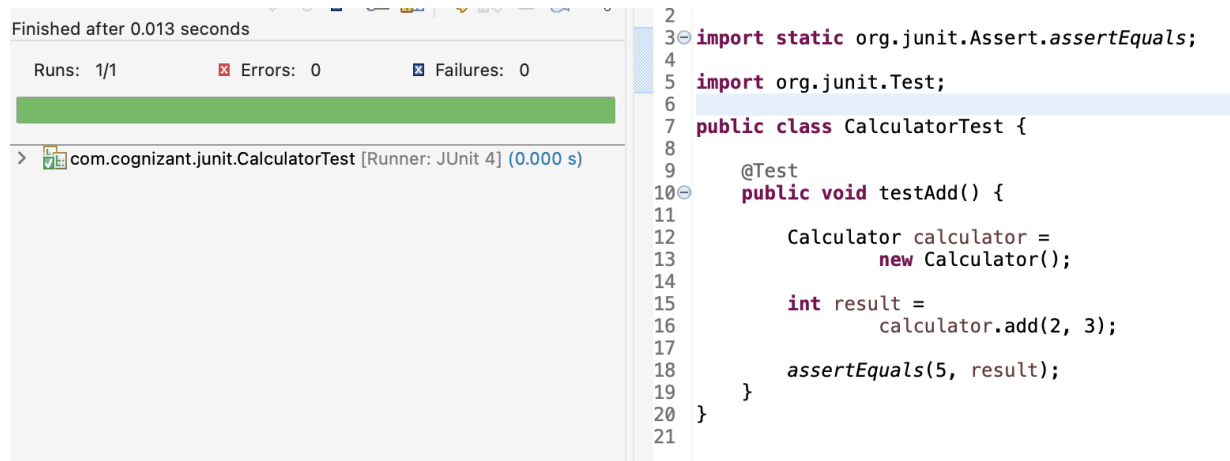


Cognizant Digital Nurture-5.0:

1.Junit Basic Testing1-

EXERCISE 1:



The screenshot shows an IDE with a test runner window on the left and a code editor on the right. The test runner window displays "Finished after 0.013 seconds", "Runs: 1/1", "Errors: 0", and "Failures: 0". Below this, a green progress bar indicates a successful run. The test name is "com.cognizant.junit.CalculatorTest [Runner: JUnit 4] (0.000 s)". The code editor shows the following Java code:

```
2
3 import static org.junit.Assert.assertEquals;
4
5 import org.junit.Test;
6
7 public class CalculatorTest {
8
9     @Test
10    public void testAdd() {
11
12        Calculator calculator =
13            new Calculator();
14
15        int result =
16            calculator.add(2, 3);
17
18        assertEquals(5, result);
19    }
20 }
21
```

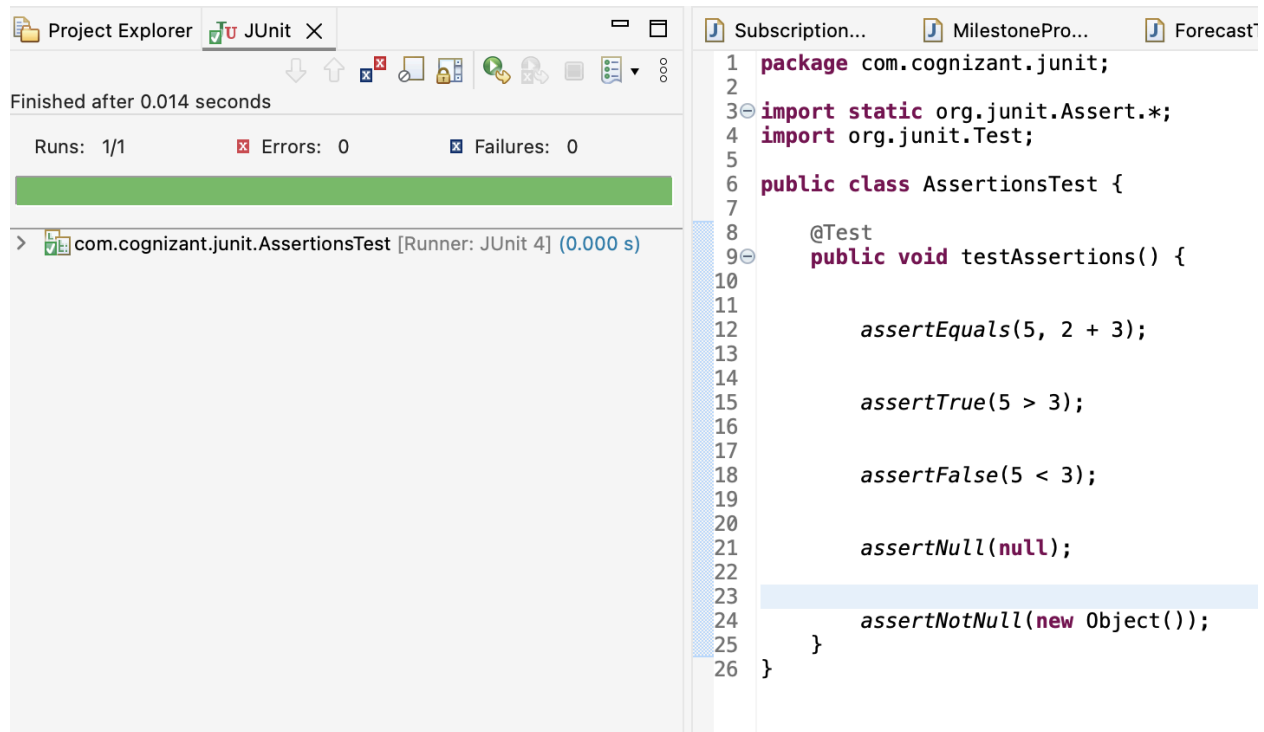
```
package com.cognizant.junit;
```

```
public class Calculator {
```

```
    public int add(int a, int b) {
        return a + b;
    }
```

```
}
```

EXERCISE3:



Project Explorer JUnit X

Finished after 0.014 seconds

Runs: 1/1 Errors: 0 Failures: 0

> com.cognizant.junit.AssertionsTest [Runner: JUnit 4] (0.000 s)

```
1 package com.cognizant.junit;
2
3 import static org.junit.Assert.*;
4 import org.junit.Test;
5
6 public class AssertionsTest {
7
8     @Test
9     public void testAssertions() {
10
11         assertEquals(5, 2 + 3);
12
13         assertTrue(5 > 3);
14
15         assertFalse(5 < 3);
16
17         assertNull(null);
18
19         assertNotNull(new Object());
20
21     }
22 }
```

EXERCISE 4:



1 package com.cognizant.junit;
2
3 import static org.junit.Assert.assertEquals;
4
5 import org.junit.After;
6 import org.junit.Before;
7 import org.junit.Test;
8
9 public class CalculatorAAATest {
10
11 private Calculator calculator;
12 @Before
13 public void setUp() {
14 calculator = new Calculator();
15 System.out.println("Setup Executed");
16 }
17
18 @After
19 public void tearDown() {
20 System.out.println("Teardown Executed");
21 }
22
23 @Test
24 public void testAddition() {
25 int a = 10;
26 int b = 20;
27 int result = calculator.add(a, b);
28 assertEquals(30, result);
29 }
30
31 @Test
32 public void testSubtraction() {
33 int a = 20;
34 int b = 5;
35 int result = calculator.subtract(a, b);
36 assertEquals(15, result);
37 }
38 }

Problems Javadoc Declaration Console X

<terminated> CalculatorAAATest [JUnit] /Library/Java/JavaVirtualMachines/jdk-25.jdk

Setup Executed
Teardown Executed
Setup Executed
Teardown Executed

Finished after 0.015 seconds

Runs: 2/2

Errors: 0

Failures: 0

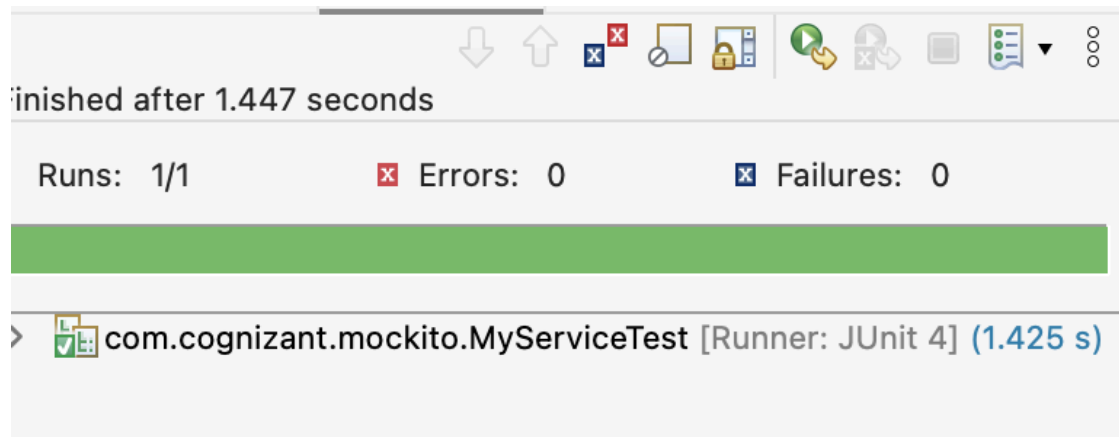
>  com.cognizant.junit.CalculatorAAATest [Runner: JUnit 4] (0.000 s)

```
1 package com.cognizant.junit;
2
3 public class Calculator {
4
5     public int add(int a, int b) {
6         return a + b;
7     }
8
9     public int subtract(int a, int b) {
10        return a - b;
11    }
12 }
```

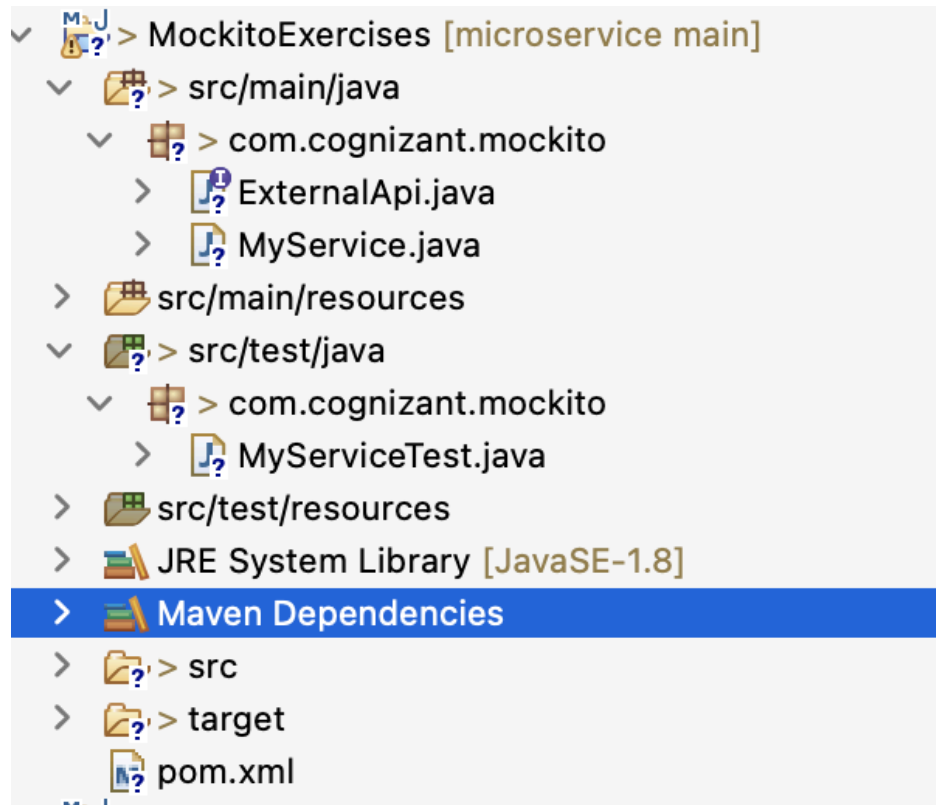
2.Mockito:

EXERCISE 1:

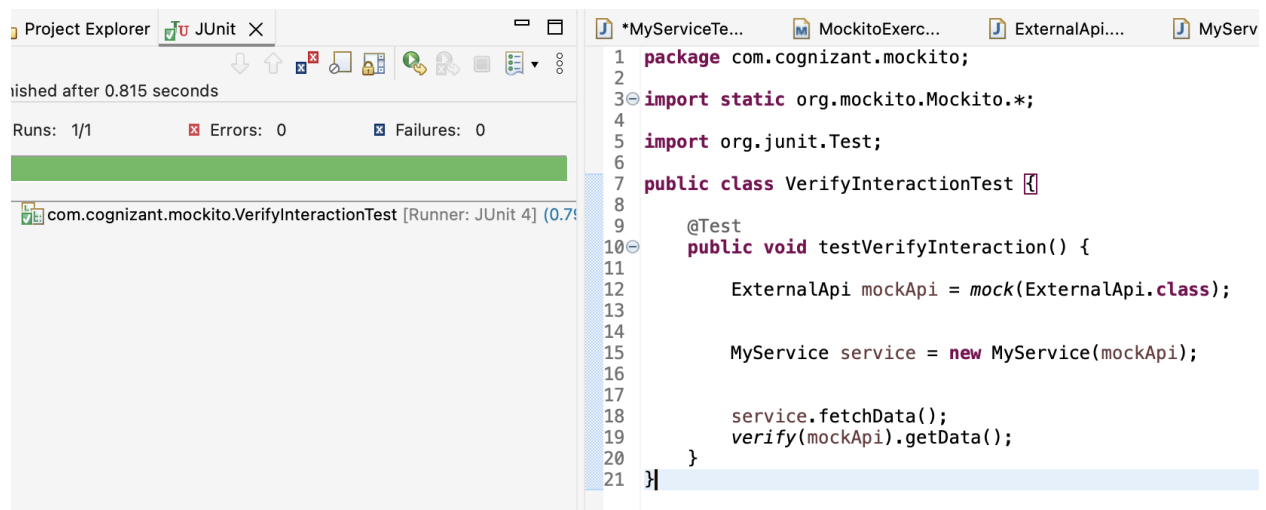
```
*MyService.java *MyServicee... MockitoExerc...
1 package com.cognizant.mockito;
2
3 import static org.junit.Assert.assertEquals;
4 import static org.mockito.Mockito.*;
5
6 import org.junit.Test;
7
8 public class MyServiceTest {
9
10     @Test
11     public void testExternalApi() {
12
13         ExternalApi mockApi =
14             mock(ExternalApi.class);
15
16         when(mockApi.getData())
17             .thenReturn("Mock Data");
18
19         MyService service =
20             new MyService(mockApi);
21
22         String result =
23             service.fetchData();
24
25         assertEquals(
26             "Mock Data",
27             result
28         );
29     }
30 }
```



```
1 package com.cognizant.mockito;
2
3 public interface ExternalApi {
4     String getData();
5 }
6
7
8 package com.cognizant.mockito;
9
10 public class MyService {
11     private ExternalApi api;
12
13     public MyService(ExternalApi api) {
14         this.api = api;
15     }
16
17     public String fetchData() {
18         return api.getData();
19     }
20 }
```

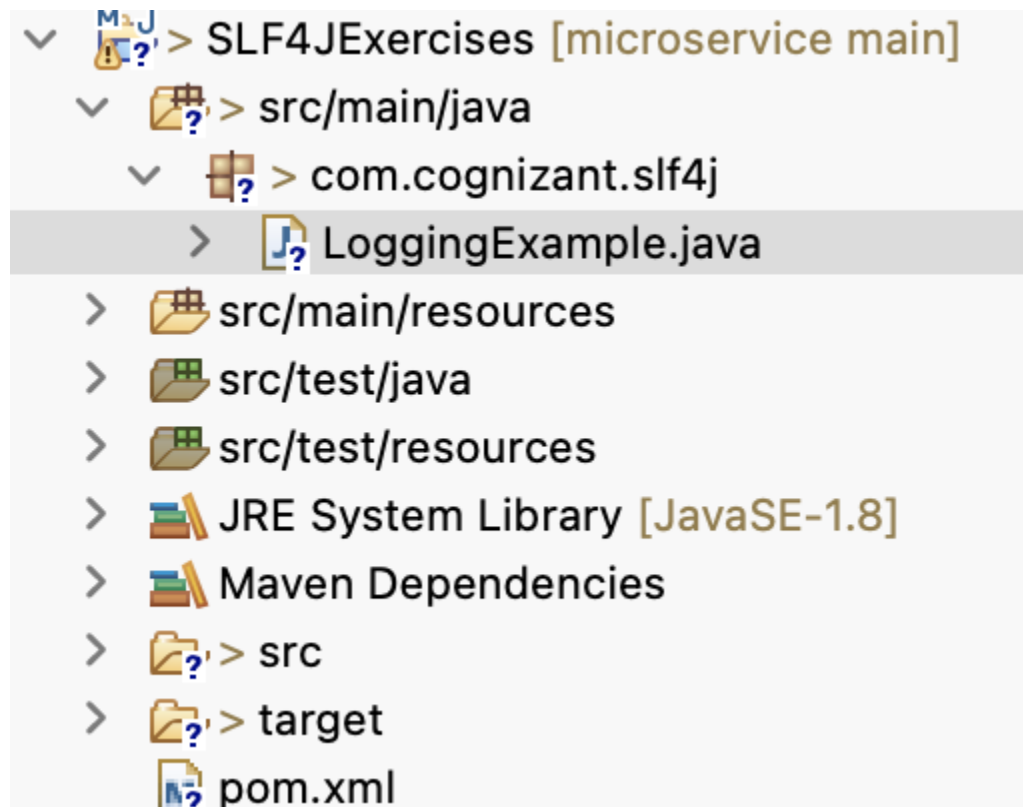


EXERCISE 2:



6.SLF4J:

EXERCISE 1-



```
1 package com.cognizant.slf4j;
2
3 import org.slf4j.Logger;
4 import org.slf4j.LoggerFactory;
5
6 public class LoggingExample {
7
8     private static final Logger logger =
9         LoggerFactory.getLogger(LoggingExample.class);
10
11     public static void main(String[] args) {
12
13         logger.error("This is an error message");
14
15         logger.warn("This is a warning message");
16     }
17 }
```

```
23:26:02.952 [main] ERROR com.cognizant.slf4j.LoggingExample - This is an error message
23:26:02.954 [main] WARN com.cognizant.slf4j.LoggingExample - This is a warning message
```